

Ashwin Conrad

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Mechanical engineering student focused on vehicle systems, manufacturing, testing, and build-ready design. Experience in Formula EV low-voltage integration, industrial control-panel QA, engineering analysis, and Python automation.

EDUCATION

University of Alberta | Mechanical Engineering - Edmonton, AB | **2023 - Present**

- Mechanical Engineering student; Formula EV Low Voltage Electronics Lead and Engineering Student Engagement Leader.

West Island College | High School - Calgary | **2020-2023**

- Played on the Basketball, Track, and Rugby teams along side Club Basketball and volunteer coaching. Also an active member in the Diversity Equity and Inclusion Club at school supporting a multicultural community.

CORE CAPABILITIES

Vehicle Systems Integration	Control Panel Assembly & QA	Manufacturing & Test
CAD & Engineering Drawings	Python & Engineering Data	Cross-functional Coordination

EXPERIENCE

Integrity & Reliability Engineering Student 2025 – 2026

AltaGas - Calgary, AB

- Built Python, Microsoft Fabric, and Power BI workflows for corrosion monitoring and inspection planning.
- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers.
- Screened results against OEM limits to identify monitoring gaps and support risk-based recommendations.
- Supported asset-lifecycle work across multi-facility operations.

Control Panel Technician Summer 2023 & 2024

Spartan Controls - Calgary, AB

- Built, wired, loaded, and checked 30+ industrial control panels from drawings and electrical schematics.
- Performed pre-delivery quality checks and documented deficiency resolution before customer delivery.
- Processed 250+ PLC software updates in under half the allocated time while supporting production.
- Supported repeatable assembly, quality, and delivery work in a production environment.

LEADERSHIP

Low Voltage Electronics Lead Winter 2025 - Present

Formula EV Racing - University of Alberta

- Lead Formula UA27 low-voltage planning and integration across hardware, safety interfaces, sensors, power distribution, and vehicle packaging.

Engineering Student Engagement Leader 2024-2025

University of Alberta - Edmonton, AB

- Served as an Engineering Student Engagement Leader and received the Undergraduate Student Leadership Award in 2024-2025.

COMMUNITY INVOLVEMENT

Basketball Coach

CYDC

- Youth coaching and team development.

RECOGNITION

Undergraduate Student Leadership Award 2024-2025

- Received the Undergraduate Student Leadership Award during the 2024-2025 academic year.

Leadership in Engineering Scholarship 2023-2024

- Recipient of the Leadership in Engineering Scholarship for 2023-2024.

APEGA Science Olympics Medalist: Silver 2021-2023

- Silver medalist in APEGA Science Olympics during 2021-2023.

SELECTED PROJECT PORTFOLIO

Brazed Aluminum Heat-Exchanger Lifecycle Analysis 2025 – 2026

AltaGas - Python, PI Historian Excel add-in

- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers.
- Assessed fatigue risk and monitoring gaps against available operating history.
- Screened lifecycle evidence against OEM thermal and cycling limits.
- Summarized findings into inspection and operating-awareness recommendations.

Inspection Scope Generation Tool 2025 – 2026

AltaGas - Python, openpyxl, python-docx

- Developed a Python tool for equipment-specific inspection-scope generation.
- Generated 500+ scopes from database exports.
- Used openpyxl and python-docx to standardize Excel-to-Word output.
- Replaced a manual Power Query and Power BI export workflow.

Formula EV Electrical Systems Winter 2025 – Present

Formula EV Racing - Low-voltage electrical systems

- Planned the Formula UA27 low-voltage system around the vehicle package.
- Coordinated power distribution and sensor integration work.
- Supported safety-interface planning with mechanical and software teammates.
- Considered accessibility and serviceability alongside electrical function.

Leather Tool Carrier 2024

Spartan Controls - Shop workflow tooling

- Designed a leather tool carrier for common hand tools used during panel assembly.
- Built the carrier as a durable, hands-on shop aid.
- Kept frequently used tools organized and accessible.
- Turned a personal workflow problem into support for faster repeated work.

TECHNICAL SKILLS

CAD & DESIGN	SolidWorks, AutoCAD & 2D laser cutting, Engineering drawings, Mechanical prototyping, 3D modelling and printing
VEHICLE & ELECTRICAL	Low-voltage systems, Power distribution, Sensor integration, Electrical schematics, Panel assembly, Pre-delivery QA
DATA & AUTOMATION	Python, Excel data analysis, Power BI, Microsoft Fabric, Power Query, openpyxl, python-docx
ENGINEERING SOFTWARE	Aspen HYSYS, Maximo, JDE
FABRICATION & QA	Engineering drawings, Mechanical prototyping, Fabrication and materials, Panel assembly

Expanded case studies and technical work: ashwin-conrad.github.io